Cognivis Test Document

Smart Campus Energy Management System (SCEMS)

Document Type: Technical System Specification
Version: 1.0
Purpose: Sample document for testing Cognivis document retrieval, question answering, evidence citation, workflow generation, architecture generation, and knowledge extraction.

1. Executive Summary

The Smart Campus Energy Management System (SCEMS) is a campus-scale platform designed to monitor electricity consumption, identify abnormal energy usage, predict short-term demand, and recommend energy-saving actions. The system collects data from smart meters and environmental sensors, processes the incoming data through an edge gateway, stores historical readings, and applies machine learning models for forecasting and anomaly detection.

The platform provides a web dashboard for facility managers. It displays current consumption, daily and monthly trends, predicted demand, detected anomalies, and recommended actions. The target is to reduce avoidable energy consumption without affecting normal campus operations.

2. System Objectives

The primary objectives of SCEMS are:

• Monitor electricity consumption across buildings in near real time.
• Forecast electricity demand for the next 24 hours.
• Detect unusual consumption patterns and possible equipment faults.
• Provide evidence-based energy-saving recommendations.
• Give facility managers a centralized visual dashboard.

3. High-Level Architecture

The system contains five major layers: sensing, edge processing, data services, intelligence, and presentation. Smart meters and environmental sensors send readings to an edge gateway. The gateway validates the data and forwards it to the backend through an MQTT message broker. The backend stores raw and processed readings in a PostgreSQL database. The analytics service uses a forecasting model and an anomaly detection model. The React dashboard retrieves processed results through REST APIs.

Architecture Components

1. Smart Meters: Measure voltage, current, power and cumulative energy usage.
2. Environmental Sensors: Measure temperature, humidity and occupancy-related signals.
3. Edge Gateway: Performs validation, timestamp normalization and temporary buffering.
4. MQTT Broker: Transfers sensor messages between edge devices and backend services.
5. Data Service: Stores readings and exposes REST endpoints.
6. Forecasting Service: Predicts electricity demand using historical readings and contextual features.
7. Anomaly Service: Identifies unusual consumption patterns using an isolation-based model.
8. Recommendation Engine: Converts predictions and anomalies into actionable suggestions.
9. Web Dashboard: Presents metrics, charts, alerts and recommendations to facility managers.

4. Data Flow

A typical reading follows this sequence: Smart Meter → Edge Gateway → MQTT Broker → Data Service → PostgreSQL → Analytics Services → Recommendation Engine → REST API → React Dashboard. If a reading fails validation, the gateway places it in a temporary buffer and retries transmission. The dashboard does not directly access the database; all application data is retrieved through the backend API.

5. Processing Workflow

Every incoming sensor message is first validated. Invalid values are rejected and logged. Valid values are normalized into a common timestamp and measurement format. The processed reading is stored in PostgreSQL. The forecasting service periodically creates a 24-hour demand forecast. The anomaly service evaluates recent readings against learned consumption patterns. When an anomaly is detected, the recommendation engine checks the building, time, recent history and environmental conditions before generating a recommendation.

6. Machine Learning Components

SCEMS uses two primary machine learning components. The demand forecasting component predicts electricity consumption for the following 24 hours. It uses recent consumption, hour of day, day of week, temperature and occupancy level as input features. The anomaly detection component identifies observations that differ substantially from normal building-level patterns.

Model Inputs

Feature	Example	Used By
Power consumption	18.7 kW	Forecasting, anomaly detection
Temperature	29.4 °C	Forecasting, recommendation
Humidity	68%	Recommendation
Occupancy	142 people	Forecasting, recommendation
Hour of day	14:00	Forecasting
Day of week	Tuesday	Forecasting

7. Recommendation Logic

Recommendations are generated only when the system has sufficient supporting evidence. For example, if a building's electricity usage is 35% above its expected level for the same time period and occupancy has not increased, the system may recommend checking HVAC schedules or non-essential equipment. Recommendations include the observed condition, supporting metrics, possible cause, and suggested action.

8. Dashboard Modules

The web dashboard contains six primary modules: Overview, Live Consumption, Forecast, Anomalies, Recommendations, and Building Comparison. The Overview page displays total campus consumption, current demand, forecasted peak demand and active alerts. The Building Comparison page allows facility managers to compare daily and monthly consumption across buildings.

9. Security and Reliability

The backend uses authenticated API access. User roles determine whether a user can view data, manage sensors, or configure recommendations. Sensor messages are timestamped and validated. The system keeps an audit record of rejected readings and recommendation decisions. Temporary edge buffering is used to prevent short network interruptions from causing immediate data loss.

10. Known Limitations

The first version does not control physical equipment automatically. Recommendations require human approval before any operational action is taken. Forecast accuracy can decrease when historical data is limited or when unusual events occur. Sensor failures can also reduce the quality of downstream analysis.

11. Expected Benefits

SCEMS is expected to help facility teams understand energy consumption, identify unusual usage earlier, plan for peak demand, and prioritize energy-saving actions. The platform is intended to support decisions with measurable evidence rather than relying only on manual inspection.

12. Test Questions for Cognivis

Use these questions after uploading this document to test Cognivis:

1. What is the main purpose of SCEMS?
2. Explain the complete data flow from the smart meter to the dashboard.
3. Generate the system architecture.
4. Generate the processing workflow.
5. What are the inputs used by the forecasting model?
6. What happens when a sensor reading fails validation?
7. What is the role of the MQTT broker?
8. Compare the forecasting and anomaly detection components.
9. What evidence is required before a recommendation is generated?
10. What are the limitations of the proposed system?

End of sample document — created specifically for testing Cognivis.